

West Bengal Council of Higher Secondary Education

HIGHER SECONDARY EXAMINATION

Modern Computer Application (COMA)

Class XII – Theory

Unit – 2 : Data Warehousing and Data Mining

Full Marks : 70

Time : 3 Hours 15 Minutes

Special credit will be given for answers which are brief and to the point. Marks will be deducted for spelling mistakes, untidiness and bad handwriting.

Figures in the margin indicate full marks for the questions.

GROUP – A (Multiple Choice Type Questions)

1. Choose the correct option for each question : [21 × 1 = 21]

1. A collection of subject-oriented, integrated, time-variant and non-volatile data is called a —

- | | |
|--------------------|-----------------|
| (a) Database | (c) Data File |
| (b) Data Warehouse | (d) Spreadsheet |

2. OLAP stands for —

- | | |
|----------------------------------|--------------------------------|
| (a) Online Analytical Processing | (c) Offline Analytical Program |
| (b) Online Access Protocol | (d) Online Application Process |

3. OLTP stands for —

- | | |
|-----------------------------------|----------------------------------|
| (a) Online Transaction Processing | (c) Offline Transaction Protocol |
| (b) Online Testing Program | (d) Online Table Processing |

4. The multidimensional structure of data in a data warehouse is called a —

- | | |
|---------------|---------------|
| (a) Data Cube | (c) Data Key |
| (b) Data Row | (d) Data Link |

5. Which of the following is NOT a feature of a data warehouse?

- (a) Subject-oriented (c) Volatile
(b) Integrated (d) Time-variant
6. The process of discovering unknown and meaningful patterns from a large data store is called —
- (a) Data Mining (c) Data Backup
(b) Data Entry (d) Data Locking
7. One of the first steps of the data mining process is —
- (a) Data Cleaning (c) Data Hiding
(b) Data Deletion (d) Data Encryption
8. The step that removes noise and missing values is called —
- (a) Data Transformation (c) Data Mining
(b) Data Cleaning (d) Data Selection
9. The step of combining data from multiple sources is called —
- (a) Data Integration (c) Data Cleaning
(b) Data Reduction (d) Data Coding
10. The step of converting data into a form suitable for analysis is called —
- (a) Data Transformation (c) Data Locking
(b) Data Deletion (d) Data Backup
11. Normalization is a strategy of which step?
- (a) Data Cleaning (c) Data Mining
(b) Data Transformation (d) Data Backup
12. OLAP is mainly used with a —
- (a) Data Warehouse (c) Printer
(b) Web Browser (d) Keyboard
13. Day-to-day business transactions are handled by —
- (a) OLTP (c) Data Mining
(b) OLAP (d) Data Cube
14. Analysing data collected in time order (e.g., daily temperature) is called —

- (a) Time Series Mining (c) Web Mining
(b) Text Mining (d) Key Mining
15. Data mining on data such as DNA or protein is called —
(a) Biological Data Mining (c) Visual Mining
(b) Audio Mining (d) Symbolic Mining
16. Extracting patterns from images or video is called —
(a) Visual Data Mining (c) Sequence Mining
(b) Text Mining (d) Statistical Mining
17. Analysing ordered events (e.g., a customer's sequential purchases) is called —
(a) Sequence Data Mining (c) Cube Mining
(b) Audio Mining (d) Row Mining
18. Analysing data using statistical methods is called —
(a) Statistical Data Mining (c) Web Mining
(b) Visual Mining (d) Key Mining
19. Which of the following is a real application of data mining?
(a) Market Basket Analysis (c) Data Typing
(b) File Printing (d) Screen Locking
20. Which is an example of data that can be mined?
(a) Relational and Transactional data (c) only broken disks
(b) only empty files (d) only empty tables
21. A future trend in data mining is —
(a) Big Data and AI-based mining (c) manual calculation
(b) paper-based accounting (d) using a typewriter

GROUP – B (Short Answer Type Questions)

2. Answer the following questions :

[14 × 1 = 14]

1. What is a Data Warehouse?
2. Write any two features of a Data Warehouse.
3. What is meant by Data Warehouse Modelling?
4. What is a Data Cube?
5. What is OLAP?

6. Write any one difference between OLTP and OLAP.
7. What is Data Mining?
8. What kinds of data can be mined? Give two examples.
9. What is Data Cleaning?
10. What is Data Transformation?
11. What is Time Series data?
12. What is Sequence Data Mining?
13. Give one example of Biological Data Mining.
14. Mention any two real applications of Data Mining.

GROUP – C (Broad / Descriptive Type Questions)**Answer any *five* of the following questions :****[5 × 7 = 35]**

1. Answer the following : **[4 + 3]**
 - (a) What is a Data Warehouse? Explain its main features.
 - (b) Briefly explain Data Warehouse Modelling.
2. Answer the following : **[3 + 4]**
 - (a) Explain what a Data Cube is, with an example.
 - (b) Write the differences between OLAP and OLTP.
3. Answer the following : **[4 + 3]**
 - (a) What is Data Mining? Explain what kinds of data and patterns can be mined.
 - (b) Briefly discuss Data Warehouse Implementation.
4. Answer the following : **[4 + 3]**
 - (a) Explain the main steps of the Data Mining process.
 - (b) Write the importance of Data Cleaning.
5. Answer the following : **[3 + 4]**
 - (a) What is Data Transformation?
 - (b) Explain the various strategies of Data Transformation.
6. Answer the following : **[4 + 3]**
 - (a) Explain Sequence Data, Time Series and Statistical Data Mining.
 - (b) Discuss Biological Data Mining with an example.
7. Answer the following : **[4 + 3]**
 - (a) Explain Visual and Audio Data Mining.
 - (b) Discuss the various applications and future trends of Data Mining.

END OF QUESTION PAPER

*Unit – 2 : Data Warehousing and Data Mining — Prepared as per WBCHSE pattern
(Group A : 21, Group B : 14, Group C : 35 = 70 marks)*